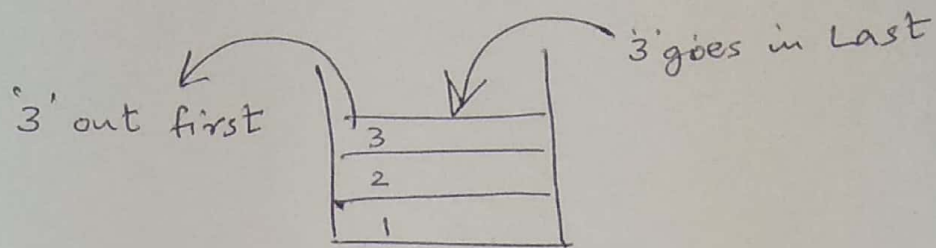


STACK

LIFO memory (Last In First Out)



Temporary memory $\rightarrow$ part of RAM

To handle STACK we have a Stack Segment (SS) Register which works with Stack Pointer (SP) Register as;

$$SS:SP \Rightarrow \{(SS) \times 10H\} + (SP)$$

'SP' locates the top of the stack memory. Data is move into and readout of Stack as '16 bits' with commands PUSH and POP respectively.

PUSH AX ; with $AX = 75A2H$

$\rightarrow$ SP decrements two bytes
e.g.; $SP = 1236$

after PUSH AX $\rightarrow$ $SP = 1234$

following little endian convention

at $SP = 1234 \rightarrow A2H$ is stored
and at $SP = 1235 \rightarrow 75H$ is stored

Process of STACK Commands

2/2

'PUSH AX'

internally this instruction is processed as

- 1- Dec SP ; $SP = SP - 1$ {where SP is ?
Dec SP ; $SP = SP - 1$ {Stack Pointer}

Here 'Dec' stands for Decrement

- 2- MOV [SP], AX

Here '[]' represents memory addressing also called direct addressing if '[]' contains a constant memory address and indirect addressing if '[]' contains a register as in our case 'SP'

step 2 can be represented in terms of Direct addressing as

MOV [1234], AX

↑
Direct address of 'SP'

'POP AX'

It is processed in reverse

- 1- ~~INC SP~~ MOV AX, [SP]

- 2- INC SP ; $SP = SP + 1$
INC SP ; $SP = SP + 1$